

R Exercises: Statistical Shape Analysis

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Notes

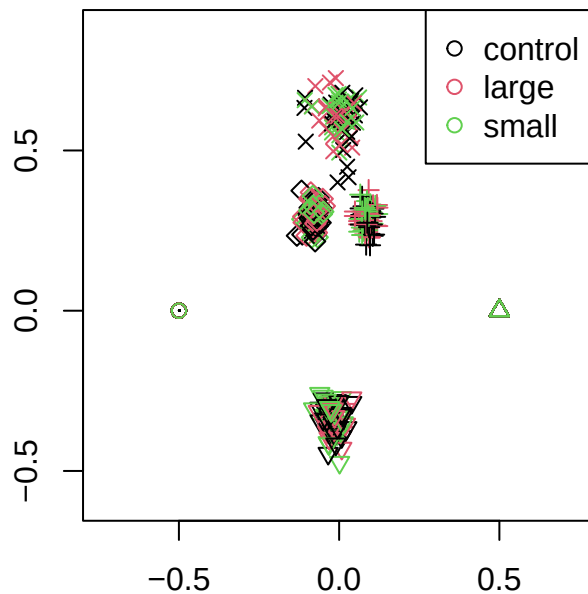
This document contains the R exercise code for the short course on Shape Analysis.

Several exercises require the R packages `shapes` and `fdasrvf`.

R Exercise 1: Bookstein shape co-ordinates

Compute and explore Bookstein shape coordinates for the mouse landmark data.

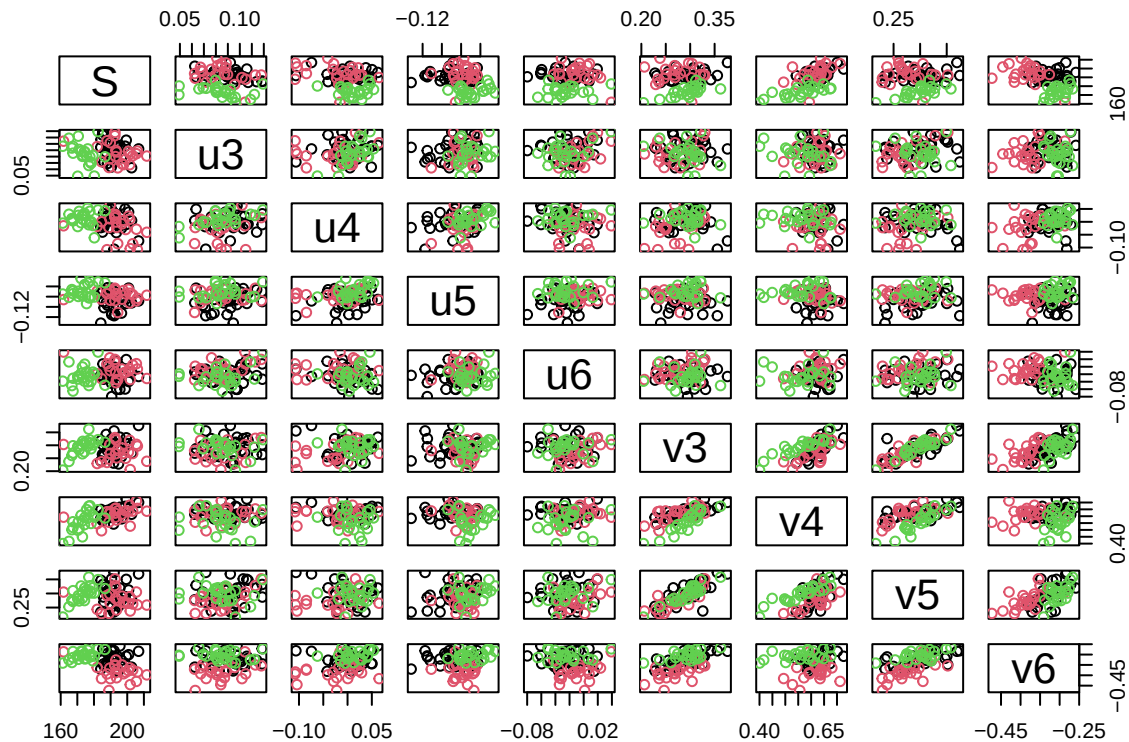
```
library("shapes")
data(mice)
u<-bookstein2d( mice$x )$bshpv
plotshapes( u , col=as.double(mice$group) , symbol=1:6)
legend("topright",c("control","large","small"),pch=1,col=c(1,2,3))
```



R Exercise 1: Bookstein shape co-ordinates — part 2

Compute and explore Bookstein shape coordinates for the mouse landmark data.

```
ub <- t(u[3:6,1,]) ; vb <- t(u[3:6,2,])
pairs( cbind( centroid.size( mice$x), ub, vb) ,
  labels=c("S","u3","u4","u5","u6","v3","v4","v5","v6"),
  col=as.double(mice$group) )
```



R Exercise 2: Procrustes shape distance

Work through pairwise Procrustes alignment and compare the manual calculation with package functions.

```
centre <- function(X) scale(X, scale = FALSE)
csize <- function(X) sqrt(sum(centre(X)^2))
preshape <- function(X) centre(X) / csize(X)
X <- mice$x[,1]
Y <- mice$x[,31]
X0 <- preshape(X)
Y0 <- preshape(Y)
sv <- svd(t(X0) %*% Y0)
R <- sv$v %*% t(sv$u)
R
```

```
##           [,1]      [,2]
## [1,] -0.9948428 -0.1014290
## [2,]  0.1014290 -0.9948428
```

```
Y_aligned <- Y0 %*% R
dP <- sqrt(sum((X0 - Y_aligned)^2))
dP
```

```
## [1] 0.09263302
```

```
print( sqrt( 2*(1-cos( riemdist( X,Y ) ))) ) #check
```

```
## [1] 0.09263302
```

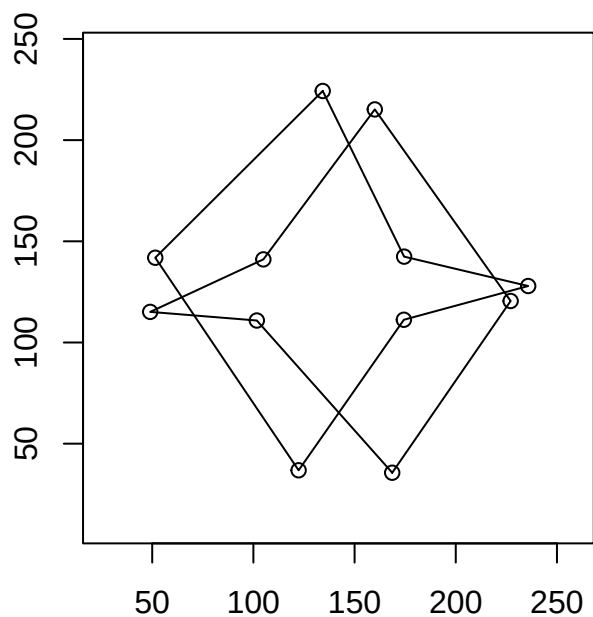
```
print( riemdist( X,Y ) ) # similar
```

```
## [1] 0.09266618
```

R Exercise 3: Pairwise Procrustes Fit

Work through pairwise Procrustes alignment and compare the manual calculation with package functions.

```
X <- mice$x[,1]
Y <- mice$x[,31]
outline <- c(1,6,2,3,4,5,1)
plotshapes( abind( X, Y) , joinline=outline )
```

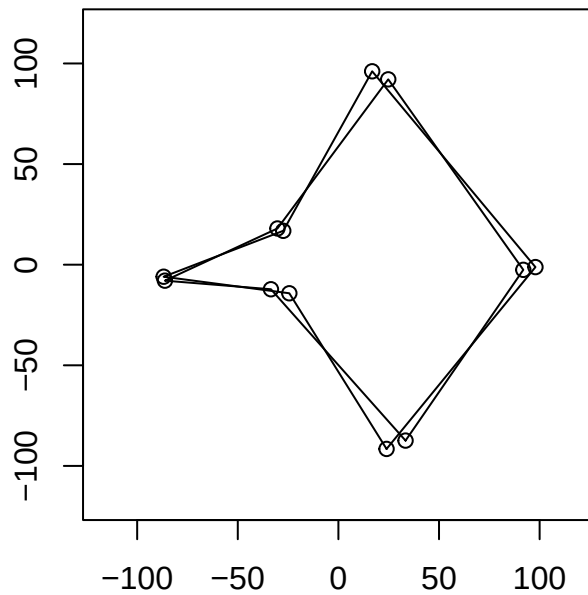


```
#match Y to X, without scaling
opal <- procOPA( X, Y, scale=FALSE)
opal
```

```
## $R
##           [,1]      [,2]
## [1,] -0.9948428 -0.1014290
## [2,]  0.1014290 -0.9948428
##
## $s
## [1] 1
##
## $Ahat
##           [,1]      [,2]
## [1,]  33.37 -87.408333
## [2,]  24.77  92.101667
## [3,] -30.29  18.001667
## [4,] -86.21  -7.938333
## [5,] -33.53 -12.198333
## [6,]  91.89 -2.558333
##
## $Bhat
##           [,1]      [,2]
## [1,]  23.93837 -91.557470
## [2,]  16.78707  96.104998
## [3,] -27.41041  16.833269
## [4,] -86.84374 -5.972611
```

```
## [5,] -24.38612 -14.210078
## [6,]  97.91482  -1.198108
##
## $OSS
## [1] 325.654
##
## $rmsd
## [1] 7.367202
```

```
plotshapes( abind( opa1$Ahat , opa1$Bhat) , joinline=outline )
```

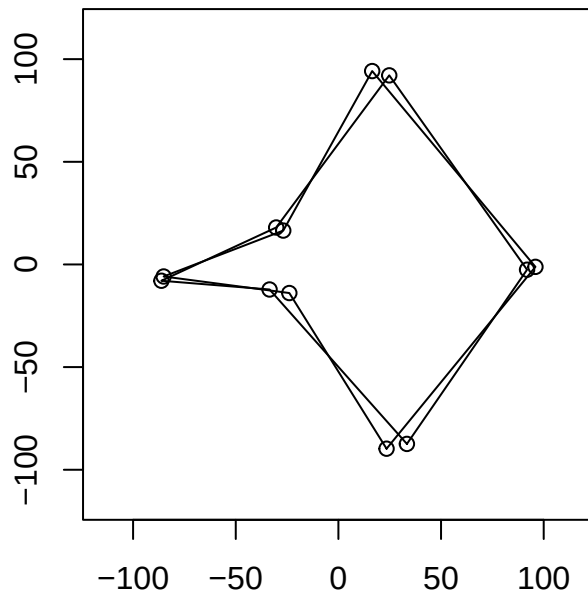


```
#match Y to X, with scaling
opa2 <- procOPA( X, Y, scale=TRUE)
opa2
```

```
## $R
##           [,1]      [,2]
## [1,] -0.9948428 -0.1014290
## [2,]  0.1014290 -0.9948428
##
## $s
## [1] 0.9801601
##
## $Ahat
##           [,1]      [,2]
## [1,]  33.37 -87.408333
## [2,]  24.77  92.101667
## [3,] -30.29  18.001667
## [4,] -86.21  -7.938333
## [5,] -33.53 -12.198333
## [6,]  91.89  -2.558333
##
## $Bhat
##           [,1]      [,2]
## [1,]  23.46344 -89.740980
## [2,]  16.45402  94.198285
```

```
## [3,] -26.86659 16.499298
## [4,] -85.12077 -5.854115
## [5,] -23.90230 -13.928151
## [6,] 95.97221 -1.174338
##
## $OSS
## [1] 310.9045
##
## $rmsd
## [1] 7.198432
```

```
plotshapes( abind( opa2$Ahat , opa2$Bhat) , joinline=outline )
```

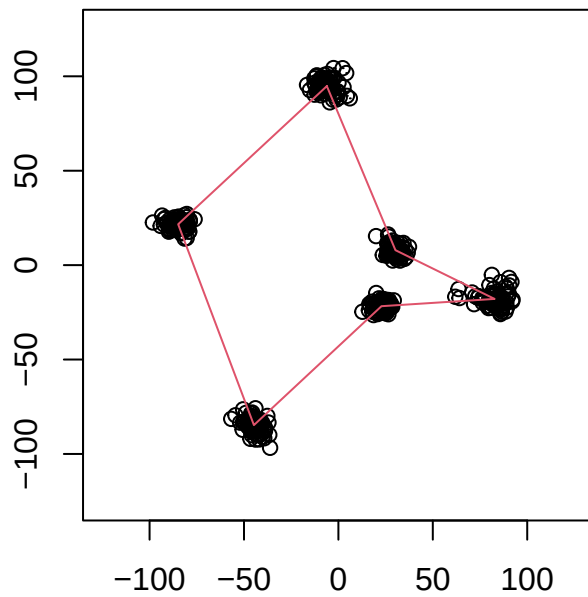


R Exercise 4: GPA with the shapes package

Use Generalized Procrustes Analysis to align configurations and estimate a mean shape.

```
data(mice)
# mice$x is k x m x n landmark array
gpa <- procGPA(mice$x, scale=TRUE)
aligned <- gpa$rotated. # aligned configurations
mean_shape <- gpa$mshape # mean shape

plotshapes(gpa$rotated)
lines( mean_shape[c(1,6,2,3,4,5,1),], col=2)
```



```
#Some DNA data
data(dna.dat)
proc <- procGPA( dna.dat , scale=FALSE)
shapes3d( proc$rotated )
```

R Exercise 5: Digit 3 data in R

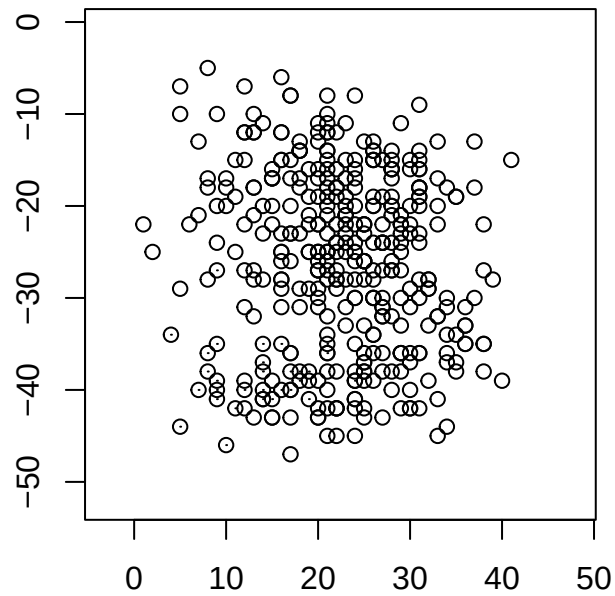
Inspect, register, and analyse the digit 3 landmark data, including the effect of an outlier.

```
library(shapes) # load the 'shapes' package (may need installing)
digit3.dat[,1:2] # Will print out the coordinates of the digit3 data
```

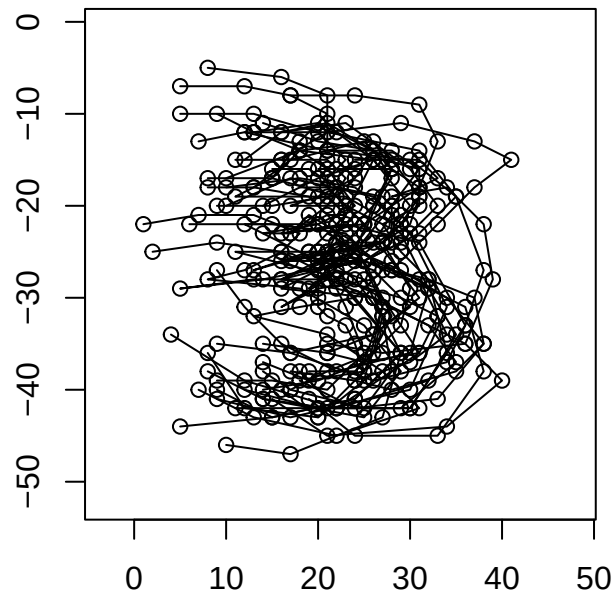
```
## , , 1
##
##      [,1] [,2]
## [1,]    9 -27
## [2,]   12 -31
## [3,]   17 -36
## [4,]   26 -39
## [5,]   34 -37
## [6,]   36 -33
## [7,]   38 -27
## [8,]   35 -19
## [9,]   30 -15
## [10,]  21 -14
## [11,]  21  -8
## [12,]  16  -6
## [13,]   8  -5
##
## , , 2
##
##      [,1] [,2]
## [1,]   17 -40
## [2,]   21 -38
## [3,]   26 -36
```

```
## [4,] 27 -32
## [5,] 25 -28
## [6,] 22 -27
## [7,] 19 -29
## [8,] 24 -25
## [9,] 26 -20
## [10,] 28 -16
## [11,] 26 -13
## [12,] 18 -14
## [13,] 15 -17
```

```
plotshapes(digit3.dat) # all 30 observations - what a mess!
```



```
plotshapes(digit3.dat, joinline=1:13) # a bit less of a mess
```



R Exercise 5: Digit 3 data in R — part 2

Inspect, register, and analyse the digit 3 landmark data, including the effect of an outlier.

```
out <- procGPA( digit3.dat , scale = TRUE)
out$rotated[,1:2]      # contains the Procrustes registered data

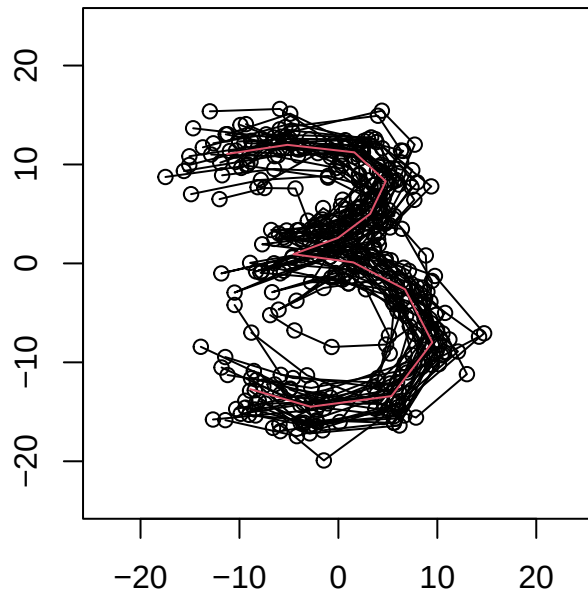
## , , 1
##
##      [,1]      [,2]
## [1,] -6.8967786 -5.2297506
## [2,] -4.4373235 -6.7753570
## [3,] -0.6849651 -8.4473150
## [4,]  4.8210398 -8.2022494
## [5,]  8.7444129 -5.4530657
## [6,]  8.9971159 -2.8672591
## [7,]  8.8337389  0.8034109
## [8,]  5.5421236  4.5187438
## [9,]  1.9978052  5.6482701
## [10,] -3.0921196  4.3183413
## [11,] -4.3403599  7.5729311
## [12,] -7.4685982  7.6175942
## [13,] -12.0160914  6.4957054
##
## , , 2
##
##      [,1]      [,2]
## [1,] -6.6105589 -16.6275271
## [2,] -1.9261216 -14.3003348
## [3,]  3.9279225 -11.9761479
## [4,]  5.1095503 -7.3007264
## [5,]  2.7823580 -2.6162891
## [6,] -0.7234568 -1.4376665
## [7,] -4.2382874 -3.7678641
## [8,]  1.6217672  0.8955363
## [9,]  3.9760070  6.7375592
## [10,]  6.3272415 11.4099754
## [11,]  3.9970439 14.9248060
## [12,] -5.3628149 13.7792416
## [13,] -8.8806508 10.2794373

plotshapes ( out$rotated , joinline=c(1:13) ) # Procrustes registered data
out$mshape      # contains the Procrustes mean shape

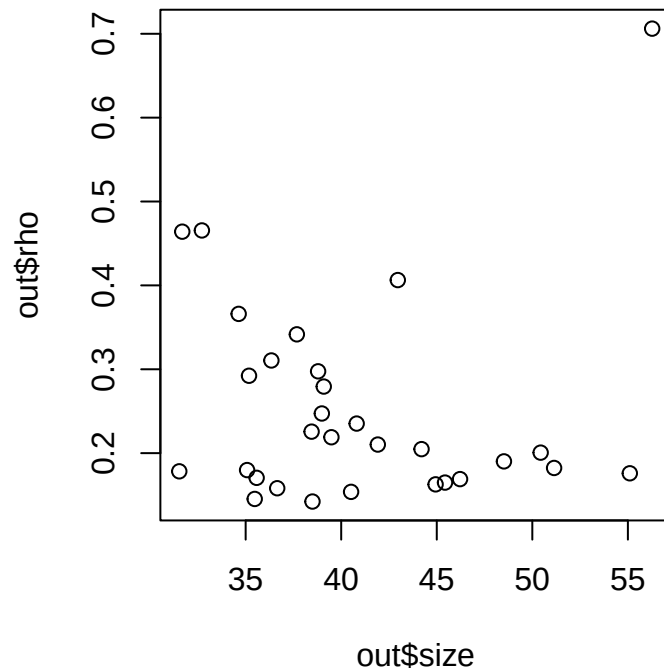
##      [,1]      [,2]
## [1,] -8.92346626 -12.6986623
## [2,] -2.80687920 -14.4874251
## [3,]  5.36447305 -13.4138524
## [4,]  9.49506198 -7.9773111
## [5,]  6.73377655 -2.6158773
## [6,]  1.51003658  0.1019867
## [7,] -4.55319977  0.9337570
## [8,] -0.03451921  2.5776875
## [9,]  3.20950982  5.0081590
## [10,]  4.78860671  8.2988427
## [11,]  1.61682677 11.2257238
```

```
## [12,] -5.15235323 11.9829801
## [13,] -11.24787380 11.0639914
```

```
lines( out$mshape, col=2 ) # adds the mean shape to the plot (in red)
```



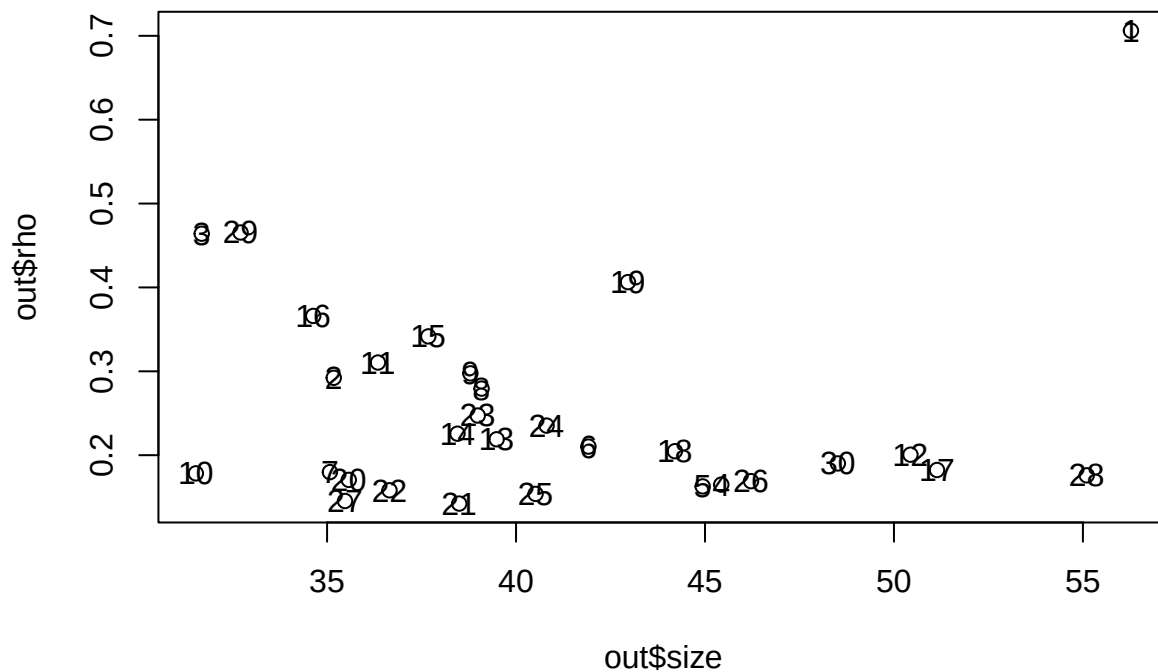
```
plot( out$size , out$rho )
```



R Exercise 5: Digit 3 (continued)

Inspect, register, and analyse the digit 3 landmark data, including the effect of an outlier.

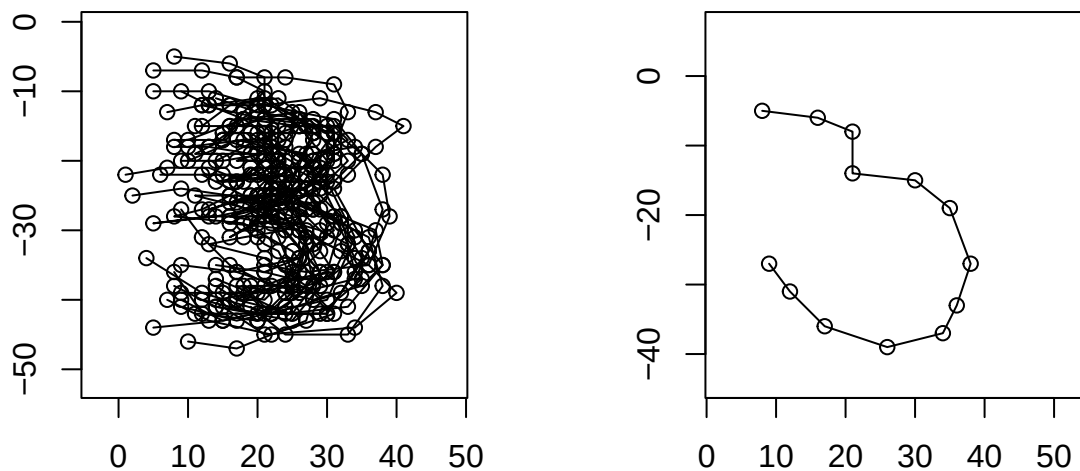
```
plot( out$size , out$rho )
text( out$size , out$rho , c(1:30) )
```



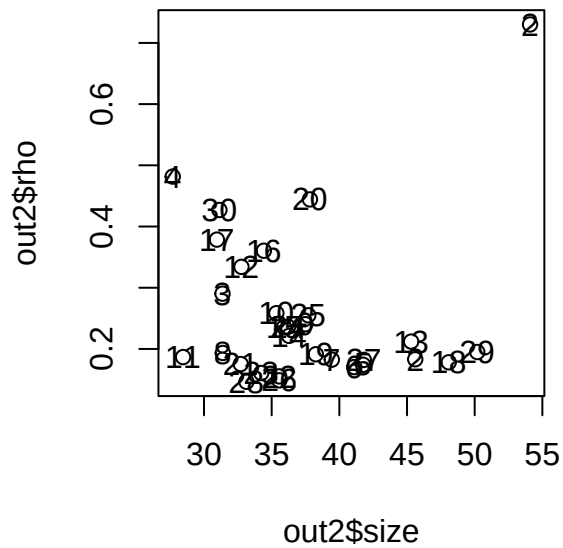
R Exercise 5: Digit 3 (continued)

Inspect, register, and analyse the digit 3 landmark data, including the effect of an outlier.

```
plotshapes( digit3.dat , digit3.dat[,1] , joinline= 1:13)
```



```
out2 <- procGPA( digit3.dat[-1,,] , scale = TRUE) #Procrustes w/o the outlier
plot( out2$size , out2$rho)
text( out2$size , out2$rho , c(2:30) )
```

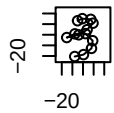


R Exercise 6: Digit 3 outlier

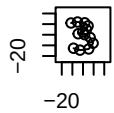
Inspect, register, and analyse the digit 3 landmark data, including the effect of an outlier.

```
out<-procGPA( digit3.dat )
shapepca(out,type="r",mag=1,joinline=1:13) # plot the effects of the first 3 PCs
```

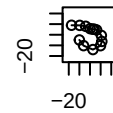
PC 1 : 43.2 %



mean - c sd

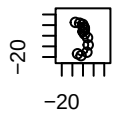


mean

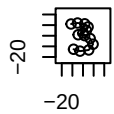


mean + c sd

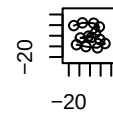
PC 2 : 15.1 %



mean - c sd

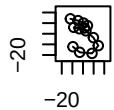


mean

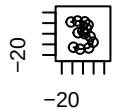


mean + c c sd

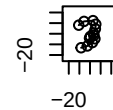
PC 3 : 14.6 %



mean - c sd



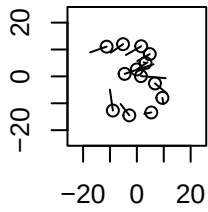
mean



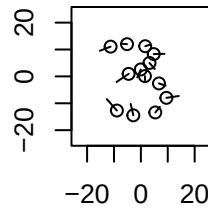
mean + c sd

```
shapepca(out,type="v",mag=1)
```

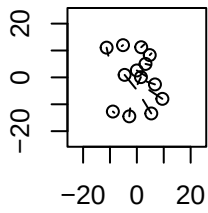
PC 1 : 43.2 %



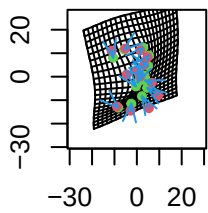
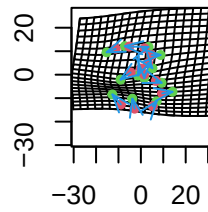
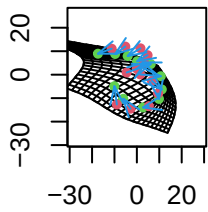
PC 2 : 15.1 %



PC 3 : 14.6 %



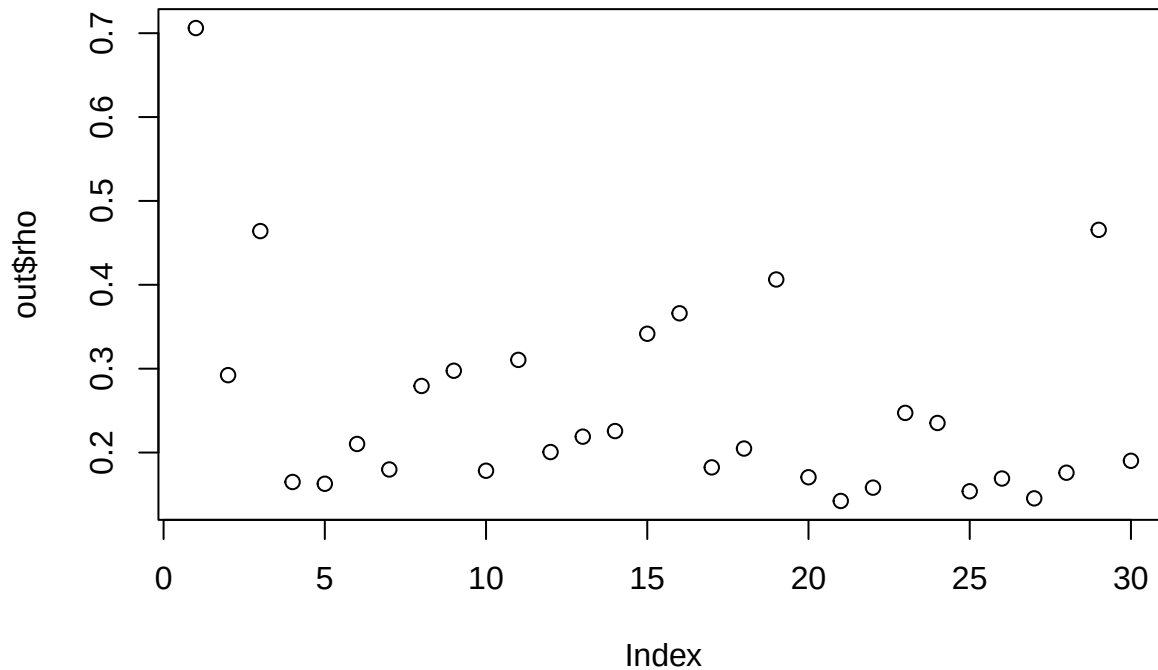
```
shapepca(out,type="g",mag=1)
```



R Exercise 6: Digit 3 outlier — part 2

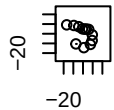
Inspect, register, and analyse the digit 3 landmark data, including the effect of an outlier.

```
plot(out$rho)
```

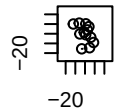


```
out2<-procGPA( digit3.dat[-1,,] )      # without outlier
shapepca(out2,type="r",mag=1)          # plot the effects of the first 3 PCs
```

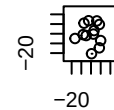
PC 1 : 43.2 %



mean - c sd

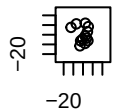


mean

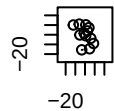


mean + c sd

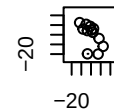
PC 2 : 17.3 %



mean - c sd

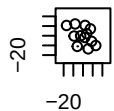


mean

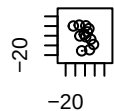


mean + c sd

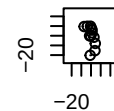
PC 3 : 13.1 %



mean - c sd



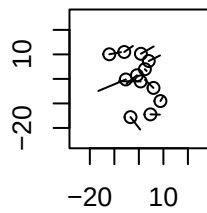
mean



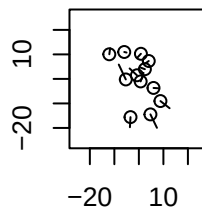
mean + c sd

```
shapepca(out2,type="v",mag=1)
```

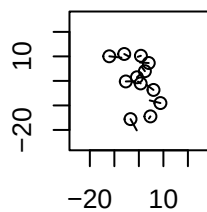
PC 1 : 43.2 %



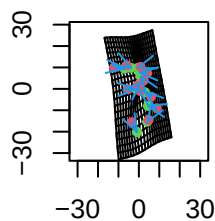
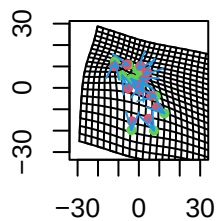
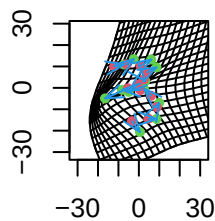
PC 2 : 17.3 %



PC 3 : 13.1 %



```
shapepca(out2,type="g",mag=1)
```



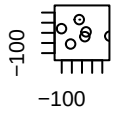
R Exercise 7: T2 mice - three groups

Analyse the three mouse groups after GPA and compare tangent-coordinate choices.

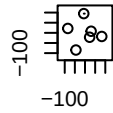
```
data(mice)
gpa <- procGPA(mice$x)
```

```
shapepca(gpa,type="r",mag=3)
```

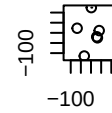
PC 1 : 36.4 %



mean - c sd

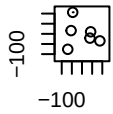


mean

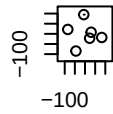


mean + c sd

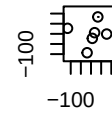
PC 2 : 28.3 %



mean - c sd

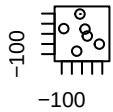


mean

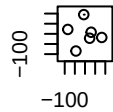


mean + c sd

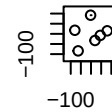
PC 3 : 14.4 %



mean - c sd



mean

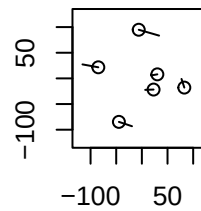
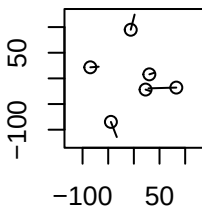


mean + c sd

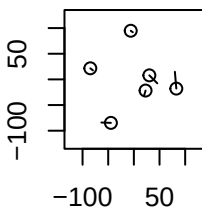
```
shapepca(gpa,type="v",mag=3)
```

PC 1 : 36.4 %

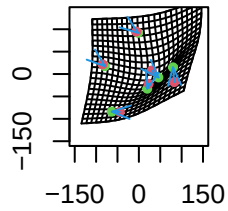
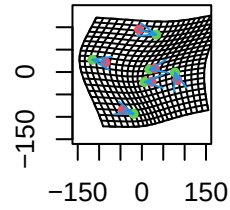
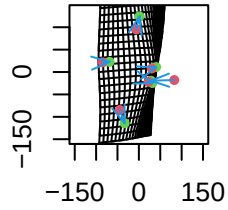
PC 2 : 28.3 %



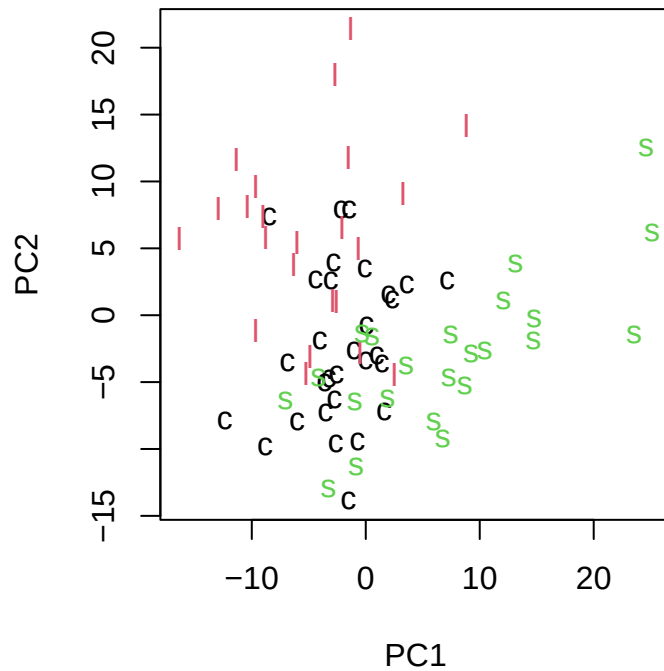
PC 3 : 14.4 %



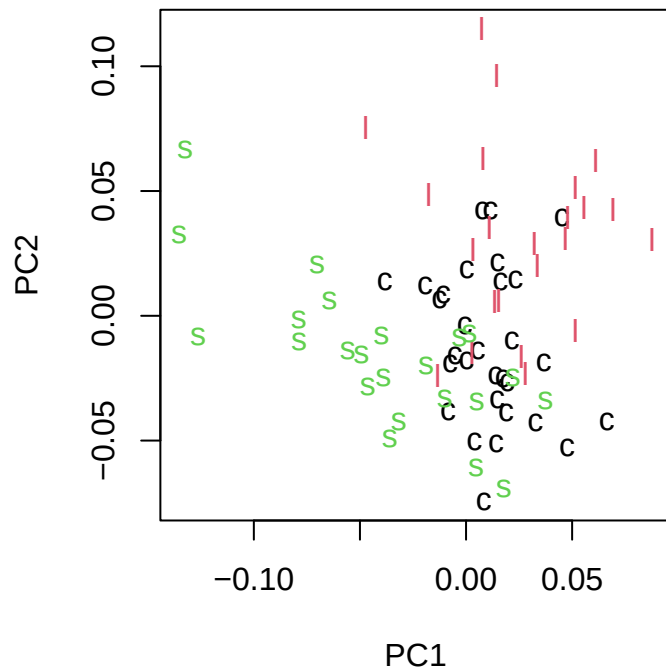

```
shapepca(gpa,type="g",mag=3)
```



```
plot(gpa$scores,col=as.factor(mice$group),pch=as.character(mice$group))
```



```
gpa2 <- procGPA(mice$x,tangentcoords="expomap")
plot(gpa2$scores,col=as.factor(mice$group),pch=as.character(mice$group))
```



R Exercise 8: 3D Monkey skulls

Read and analyse the 3D monkey skull data; this requires the external file `monkey.csv`.

```
tem <- read.csv("monkey.csv")
tem<-as.matrix(tem)
dim(tem) #Note the first column has a landmark label (number) in it.
```

```
## [1] 7 58
```

```
tem1 <- tem[,2:58]
monk <- array(tem1, c(7, 3, 19) )
```

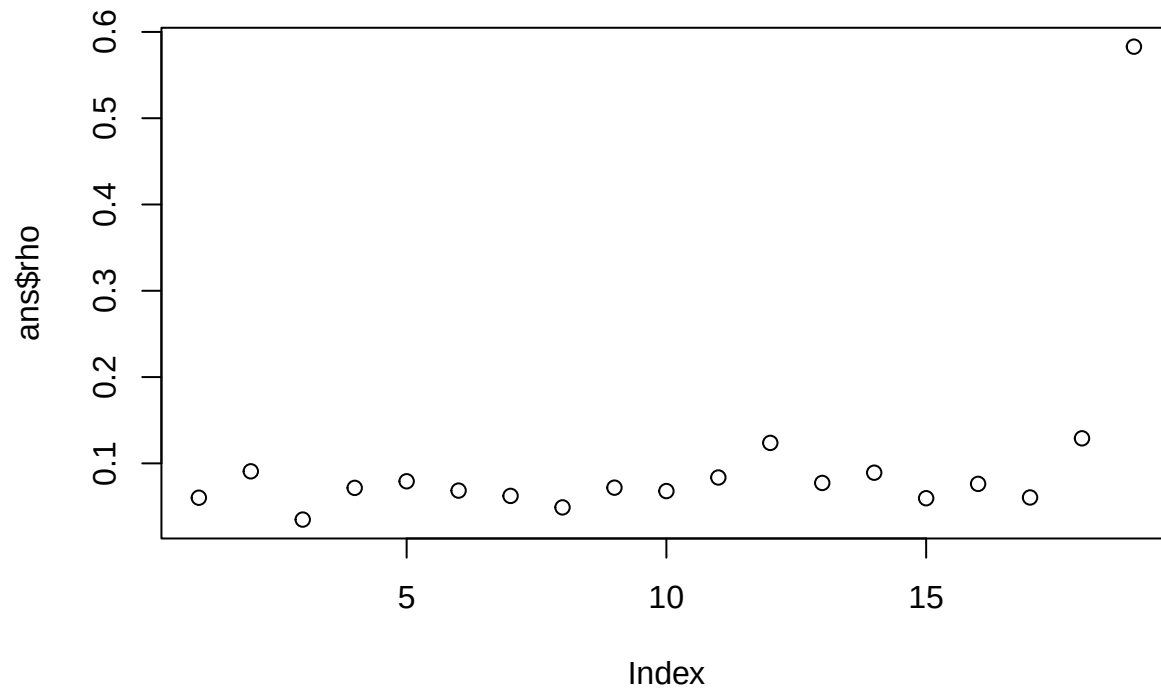
R Exercise 8: 3D Monkey skulls — part 2

Read and analyse the 3D monkey skull data; this requires the external file `monkey.csv`.

```
library(shapes)
ans <- procGPA( monk)
ans$percent
```

```
## [1] 7.614479e+01 7.506769e+00 4.866541e+00 3.564080e+00 2.554837e+00
## [6] 1.515816e+00 1.279205e+00 8.671218e-01 8.346364e-01 3.622813e-01
## [11] 2.453006e-01 1.354249e-01 8.208869e-02 4.024604e-02 8.672711e-04
## [16] 4.566864e-09 3.093958e-10 6.379167e-11 3.033642e-32
```

```
plot(ans$rho)
```



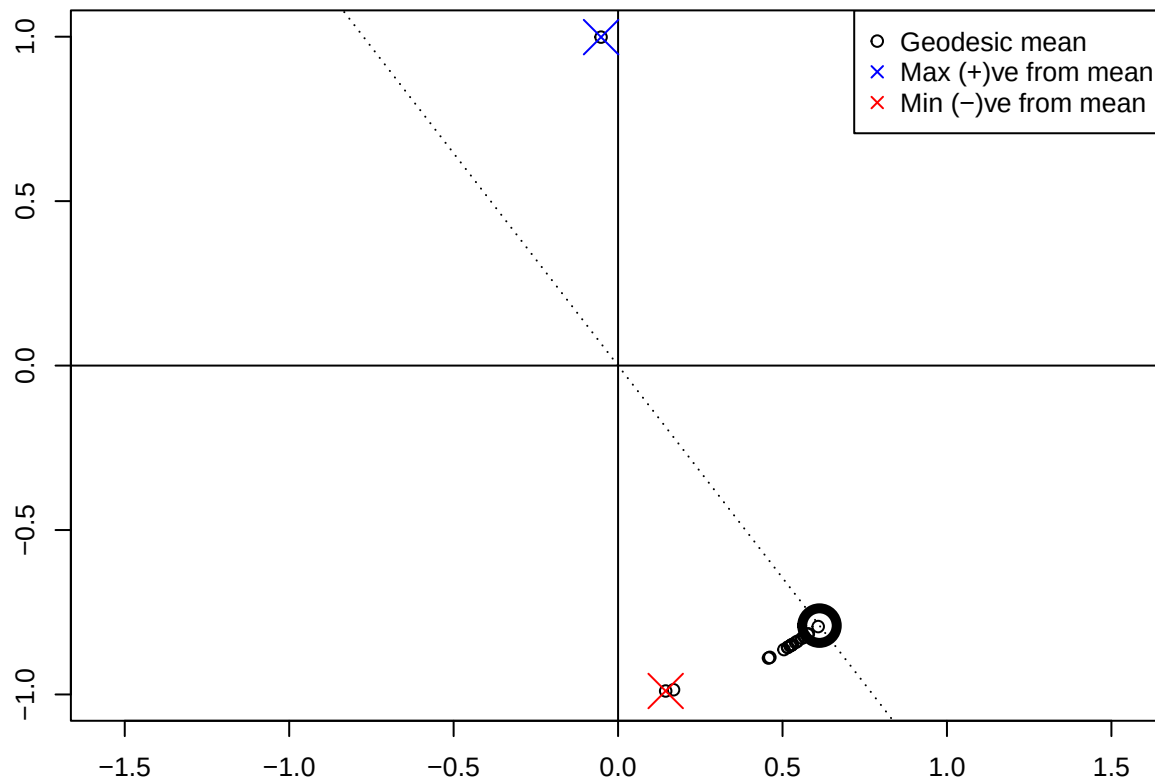
```
shapepca( ans )
```

R Exercise 8 (continued)

Run this chunk after loading any required data and packages.

```
ans2 <- pnss3d( monk , n.pc=10, sphere.type="small")
```

```
## First 10 principal components explain 99.65% of total variance.
## Message from pns() : dataset is on 10-sphere.
```



```
##
## length of BLUE from geodesic mean : 2.534342 (145 degree)
## length of RED from geodesic mean : -0.513772 (-29 degree)
##
## [1] "Radii of spheres"
## [1] 1.0000000 0.6435604 0.3492723 0.3487524 0.3405976 0.3123453 0.3123408
## [8] 0.3121839 0.3121142 0.3118684
## [1] "PNS percent explained"
## 90.74 2.66 1.93 1.41 0.81 0.73 0.47 0.45 0.31 0.14
## [1] "PCA percent explained"
## 76.79 7.33 4.8 3.51 2.51 1.49 1.23 0.85 0.82 0.31 0.19 0.11 0.04 0 0 0 0 0
```

```
ans2$percent
```

```
## [1] 90.7448655 2.6576771 1.9262463 1.4133576 0.8115303 0.7345762
## [7] 0.4662909 0.4455059 0.3144231 0.1369249
```

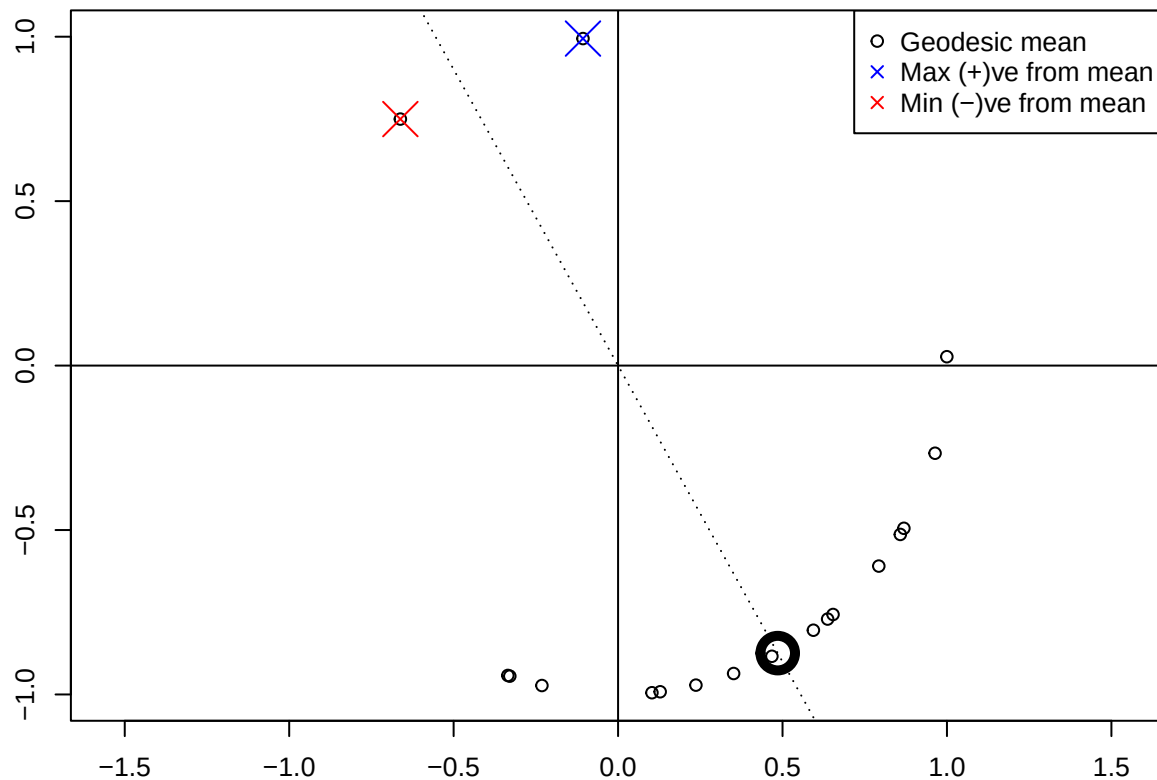
```
out <- plot3darcs( ans2, pcno=1, c=3)
```

R Exercise 8 (continued)

Run this chunk after loading any required data and packages.

```
ans3 <- pnss3d( monk[, -19] , n.pc=10, sphere.type="small")
```

```
## First 10 principal components explain 98.79% of total variance.
## Message from pns() : dataset is on 10-sphere.
```



```
##
## length of BLUE from geodesic mean : 2.741698 (157 degree)
## length of RED from geodesic mean : -2.925004 (-168 degree)
##
## [1] "Radii of spheres"
## [1] 1.00000000 0.88414915 0.13943149 0.09684907 0.09223408 0.08702585
## [7] 0.08702005 0.08349287 0.08231059 0.08051194
## [1] "PNS percent explained"
## 67.53 10.17 7.43 3.77 3.31 1.99 1.78 1.5 0.75 0.56
## [1] "PCA percent explained"
## 31.13 20.11 14.68 10.59 6.22 5.64 3.62 3.45 2.2 1.15 0.67 0.36 0.17 0 0 0 0 0
out <- plot3darcs( ans3, pcno=1, c=3)
```

R Exercise 9: DNA

Apply PCA and principal nested spheres-style analysis to the DNA dataset.

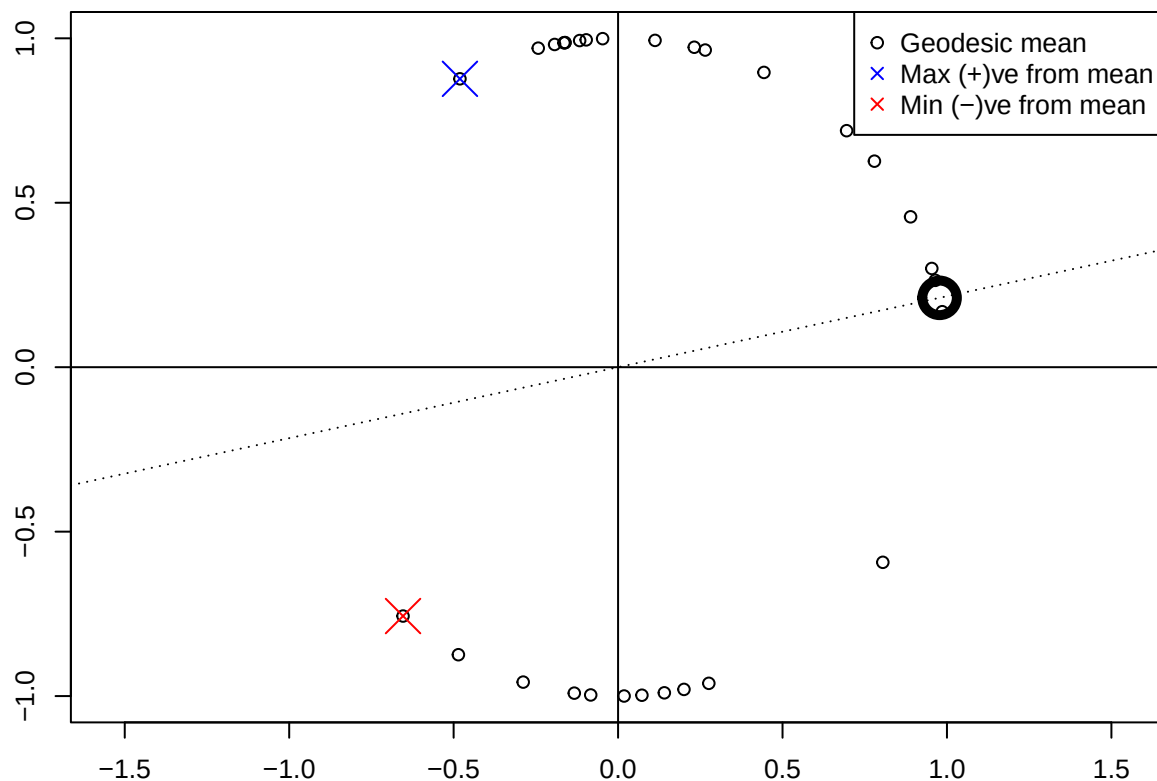
```
data(dna.dat)
proc <- procGPA( dna.dat )
proc$percent

## [1] 4.419668e+01 1.268073e+01 7.687940e+00 6.997332e+00 5.398628e+00
## [6] 3.741346e+00 2.969474e+00 2.326822e+00 2.104032e+00 1.828722e+00
## [11] 1.555211e+00 1.415392e+00 1.099450e+00 9.755162e-01 8.418280e-01
## [16] 6.688398e-01 6.427089e-01 4.468509e-01 4.126993e-01 3.820295e-01
## [21] 3.295030e-01 2.978990e-01 2.069164e-01 1.885828e-01 1.630016e-01
## [26] 1.400885e-01 1.126634e-01 1.057430e-01 8.336531e-02 8.716259e-03
```

```
shapepca(proc)
```

```
ans <- pnss3d( dna.dat , n.pc=10, sphere.type="BIC")
```

```
## First 10 principal components explain 89.94% of total variance.  
## Message from pns() : dataset is on 10-sphere.  
## .. with BIC  
## BICsm: -295.4443, BICgr: -255.7849  
## 10-sphere to 9-sphere, by SMALL sphere, BIC  
## BICsm: -94.51546, BICgr: -97.52041  
## 9-sphere to 8-sphere, by GREAT sphere, BIC  
## BICsm: -92.20934, BICgr: -95.60603  
## 8-sphere to 7-sphere, by GREAT sphere, BIC  
## BICsm: -90.46694, BICgr: -92.03279  
## 7-sphere to 6-sphere, by GREAT sphere, BIC  
## BICsm: -84.40779, BICgr: -85.32158  
## 6-sphere to 5-sphere, by GREAT sphere, BIC  
## BICsm: -82.35373, BICgr: -76.22067  
## 5-sphere to 4-sphere, by SMALL sphere, BIC  
## BICsm: -64.46519, BICgr: -60.7368  
## 4-sphere to 3-sphere, by SMALL sphere, BIC  
## BICsm: -54.83794, BICgr: -58.22728  
## 3-sphere to 2-sphere, by GREAT sphere, BIC  
## BICsm: -45.07375, BICgr: -40.68584  
## 2-sphere to 1-sphere, by SMALL sphere, BIC
```



```
##  
## length of BLUE from geodesic mean : 1.859895 (107 degree)  
## length of RED from geodesic mean : -2.495974 (-143 degree)
```

```
##
## [1] "Radii of spheres"
## [1] 1.00000000 0.06669060 0.06669060 0.06669060 0.06669060 0.06669060
## [7] 0.05845607 0.05679841 0.05679841 0.05478472
## [1] "PNS percent explained"
## 72.13 5.73 3.7 2.54 1.62 1.47 1.05 0.83 0.7 0.17
## [1] "PCA percent explained"
## 44.21 12.68 7.69 7 5.4 3.74 2.97 2.32 2.1 1.83 1.55 1.41 1.1 0.98 0.84 0.67 0.64 0.45 0.41 0.38 0.33
ans$percent

## [1] 72.1312335 5.7321424 3.6974319 2.5357671 1.6232252 1.4703298
## [7] 1.0495972 0.8318734 0.6967972 0.1703182

temp <- plot3darcs(ans, pcno=1, c=2)
```

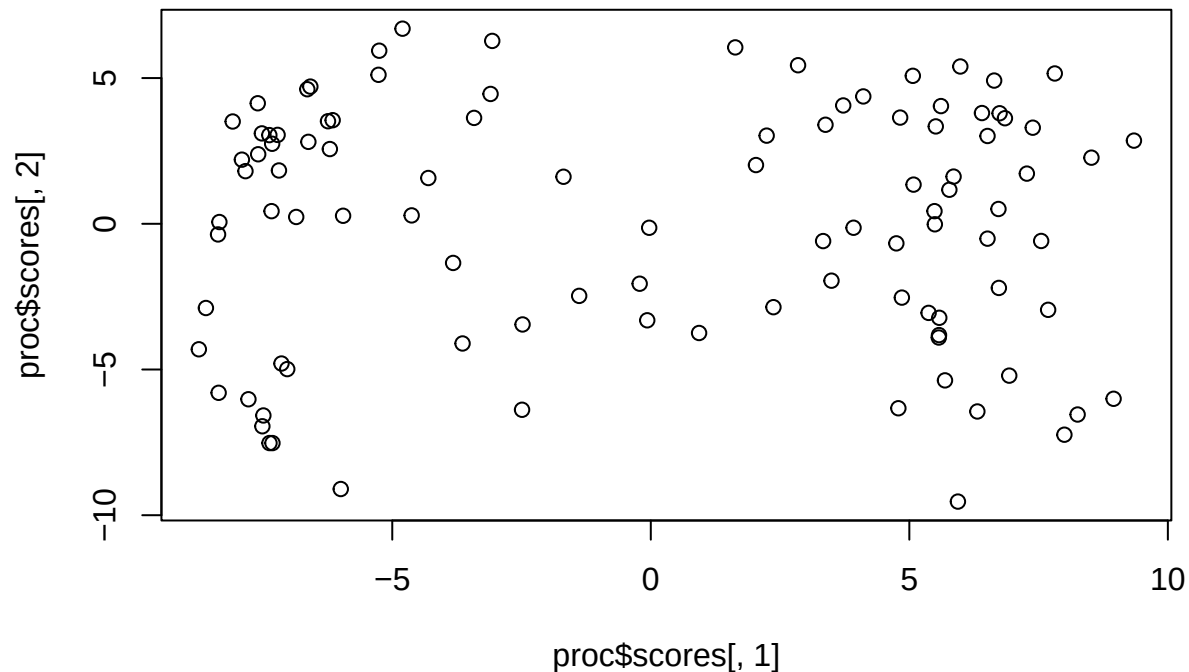
R Exercise 10: Peptides

Analyse peptide configurations; this requires the external file `peptide.Rda`.

```
load("peptide.Rda")
proc <- procGPA( z1 )
proc$percent

## [1] 3.802166e+01 1.738994e+01 1.162349e+01 7.145101e+00 5.528384e+00
## [6] 3.474069e+00 3.179235e+00 2.405219e+00 1.730728e+00 1.414570e+00
## [11] 1.341104e+00 1.085015e+00 8.585754e-01 8.216896e-01 5.728887e-01
## [16] 4.383013e-01 4.253398e-01 4.009837e-01 3.243809e-01 2.448871e-01
## [21] 2.369067e-01 1.793448e-01 1.760615e-01 1.456589e-01 1.193082e-01
## [26] 1.171824e-01 9.835969e-02 7.669551e-02 7.324771e-02 5.066645e-02
## [31] 4.630818e-02 3.688596e-02 2.522629e-02 2.492674e-02 1.943698e-02
## [36] 1.914989e-02 1.645280e-02 1.499093e-02 1.287323e-02 1.241276e-02
## [41] 9.999558e-03 8.507732e-03 7.399494e-03 6.139775e-03 4.483764e-03
## [46] 4.283915e-03 3.839710e-03 3.276469e-03 2.881905e-03 2.511762e-03
## [51] 2.453698e-03 1.993945e-03 1.739663e-03 1.307599e-03 1.177476e-03
## [56] 1.109986e-03 1.024652e-03 9.471626e-04 8.407284e-04 7.566065e-04
## [61] 6.599422e-04 5.905211e-04 5.751503e-04 5.060907e-04 4.947569e-04
## [66] 4.015546e-04 3.581816e-04 3.300481e-04 2.891129e-04 2.487178e-04
## [71] 2.262773e-04 1.923168e-04 1.730056e-04 1.437037e-04 1.088238e-04
## [76] 1.040353e-04 9.479884e-05 6.342526e-05 4.500605e-05 3.534182e-05
## [81] 2.769513e-05 1.745816e-08 2.299373e-10 4.394203e-11 2.501043e-29
## [86] 2.273756e-29 1.472679e-29

shapepca(proc)
plot(proc$scores[,1],proc$scores[,2])
```

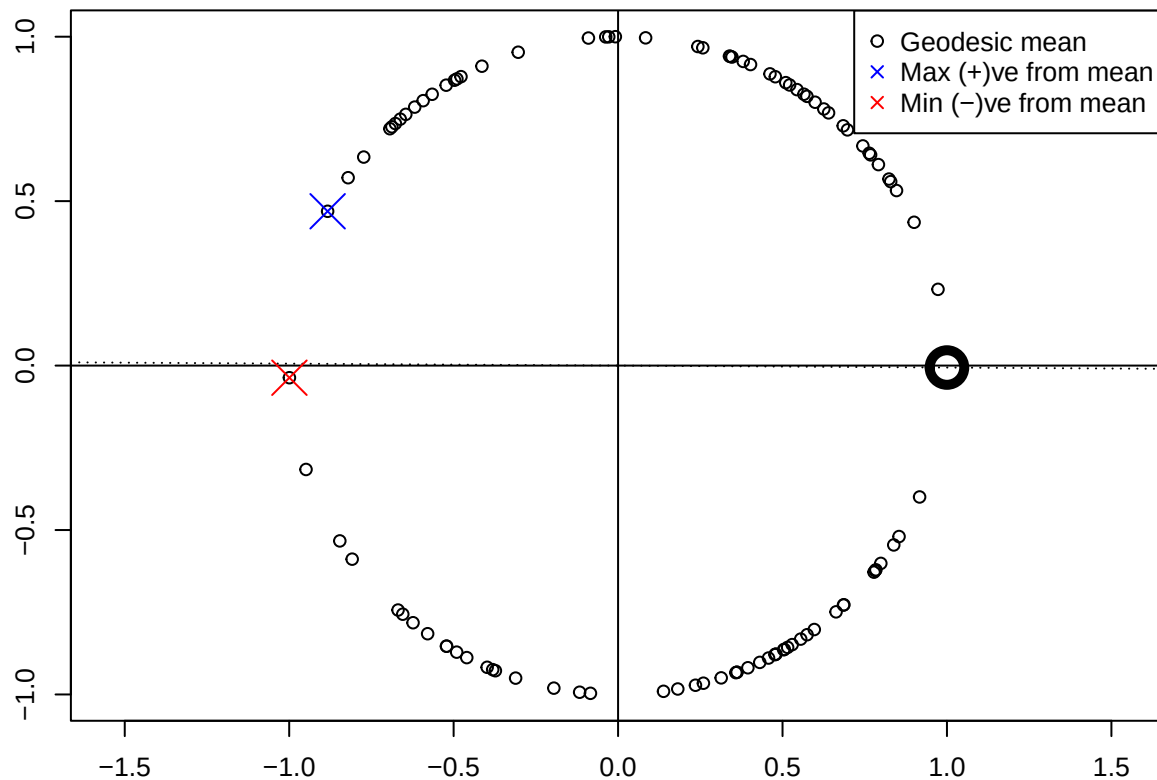


R Exercise 10: Peptides — part 2

Analyse peptide configurations; this requires the external file `peptide.Rda`.

```
ans <- pnss3d( z1 , n.pc=10, sphere.type="BIC")
```

```
## First 10 principal components explain 92.54% of total variance.
## Message from pns() : dataset is on 10-sphere.
## .. with BIC
## BICsm: -584.1837, BICgr: -555.8673
## 10-sphere to 9-sphere, by SMALL sphere, BIC
## BICsm: -421.8144, BICgr: -418.5739
## 9-sphere to 8-sphere, by SMALL sphere, BIC
## BICsm: -391.7358, BICgr: -385.5163
## 8-sphere to 7-sphere, by SMALL sphere, BIC
## BICsm: -359.2784, BICgr: -360.5301
## 7-sphere to 6-sphere, by GREAT sphere, BIC
## BICsm: -331.6859, BICgr: -331.8335
## 6-sphere to 5-sphere, by GREAT sphere, BIC
## BICsm: -331.7595, BICgr: -285.613
## 5-sphere to 4-sphere, by SMALL sphere, BIC
## BICsm: -249.673, BICgr: -250.0836
## 4-sphere to 3-sphere, by GREAT sphere, BIC
## BICsm: -198.9278, BICgr: -179.2573
## 3-sphere to 2-sphere, by SMALL sphere, BIC
## BICsm: -124.8717, BICgr: -119.259
## 2-sphere to 1-sphere, by SMALL sphere, BIC
```

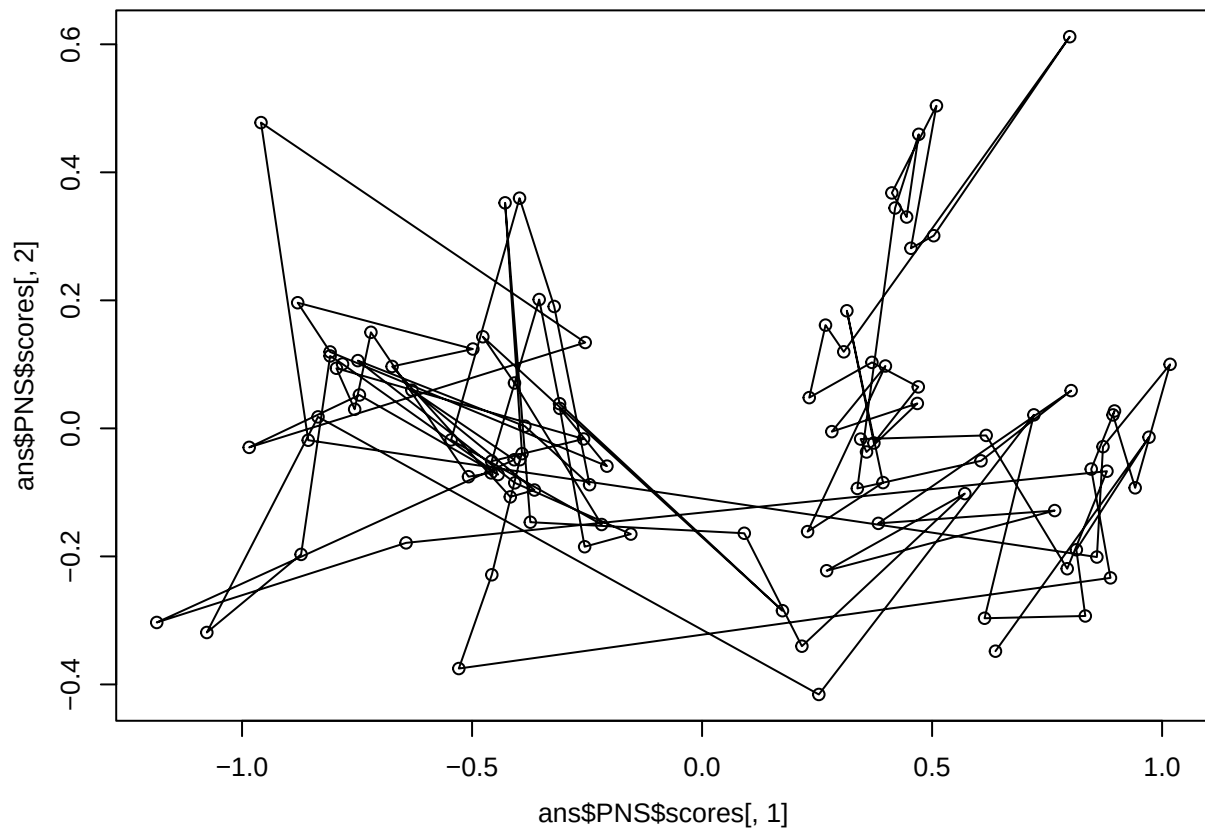



```
##
## length of BLUE from geodesic mean : 2.659249 (152 degree)
## length of RED from geodesic mean : -3.09879 (-178 degree)
##
## [1] "Radii of spheres"
## [1] 1.0000000 0.5124301 0.5014954 0.4839989 0.4839989 0.4839989 0.4189101
## [8] 0.4189101 0.3954194 0.3825568
## [1] "PNS percent explained"
## 73.18 7.88 4.03 2.41 1.3 1.3 0.93 0.67 0.49 0.35
## [1] "PCA percent explained"
## 38.78 17.73 11.66 6.88 5.53 3.53 2.81 2.47 1.74 1.41 1.13 0.98 0.85 0.76 0.57 0.43 0.42 0.36 0.25 0.1
```

```
ans$percent
```

```
## [1] 73.1785591 7.8802571 4.0275840 2.4147807 1.2990054 1.2980443
## [7] 0.9303824 0.6667790 0.4921391 0.3529049
```

```
plot(ans$PNS$scores[,1],ans$PNS$scores[,2])
lines(ans$PNS$scores[,1],ans$PNS$scores[,2],col=1:100)
```



```
tem <- plot3darcs(ans, pcno=1, c=1, type="pnss")
#compare to PCA
tem2 <- plot3darcs(ans, pcno=1, c=1, type="pca")
```

R Exercise 11: permutation test

Compare mean shapes between mouse groups using permutation and resampling tests.

```
mlarge <- mice$x[,mice$group=="l"]
msmall <- mice$x[,mice$group=="s"]
testmeanshapes( mlarge, msmall )
```

```
## Permutations - sampling without replacement: No of permutations = 1000
## 100 200 300 400 500 600 700 800 900 1000

## $H
## [1] 15.51118
##
## $H.pvalue
## [1] 0.000999001
##
## $H.table.pvalue
## [1] 1.040187e-09
##
## $G
## [1] 18.27306
##
## $G.pvalue
```

```
## [1] 0.000999001
##
## $G.table.pvalue
## [1] 0
##
## $J
## [1] 147.5658
##
## $J.pvalue
## [1] 0.000999001
##
## $J.table.pvalue
## [1] 0
```

```
resampletest( mlarge, msmall )
```

```
## Resampling...No of resamples = 200
## Bootstrap - sampling with replacement
## 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28
## $lambda
## [1] 197.7547
##
## $lambda.pvalue
## [1] 0.004975124
##
## $lambda.table.pvalue
## [1] 0
##
## $H
## [1] 15.52336
##
## $H.pvalue
## [1] 0.004975124
##
## $H.table.pvalue
## [1] 1.029129e-09
##
## $J
## [1] 147.6817
##
## $J.pvalue
## [1] 0.004975124
##
## $J.table.pvalue
## [1] 0
##
## $G
## [1] 18.33161
##
## $G.pvalue
## [1] 0.004975124
##
## $G.table.pvalue
## [1] 0
```

```
mcontrol <- mice$x[,mice$group=="c"]
testmeanshapes( mlarge, mcontrol )
```

```
## Permutations - sampling without replacement: No of permutations = 1000
## 100 200 300 400 500 600 700 800 900 1000

## $H
## [1] 6.352841
##
## $H.pvalue
## [1] 0.000999001
##
## $H.table.pvalue
## [1] 1.831439e-05
##
## $G
## [1] 8.621698
##
## $G.pvalue
## [1] 0.000999001
##
## $G.table.pvalue
## [1] 6.981216e-11
##
## $J
## [1] 59.17869
##
## $J.pvalue
## [1] 0.000999001
##
## $J.table.pvalue
## [1] 0
```

```
resampletest( mlarge, mcontrol )
```

```
## Resampling...No of resamples = 200
## Bootstrap - sampling with replacement
## 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28

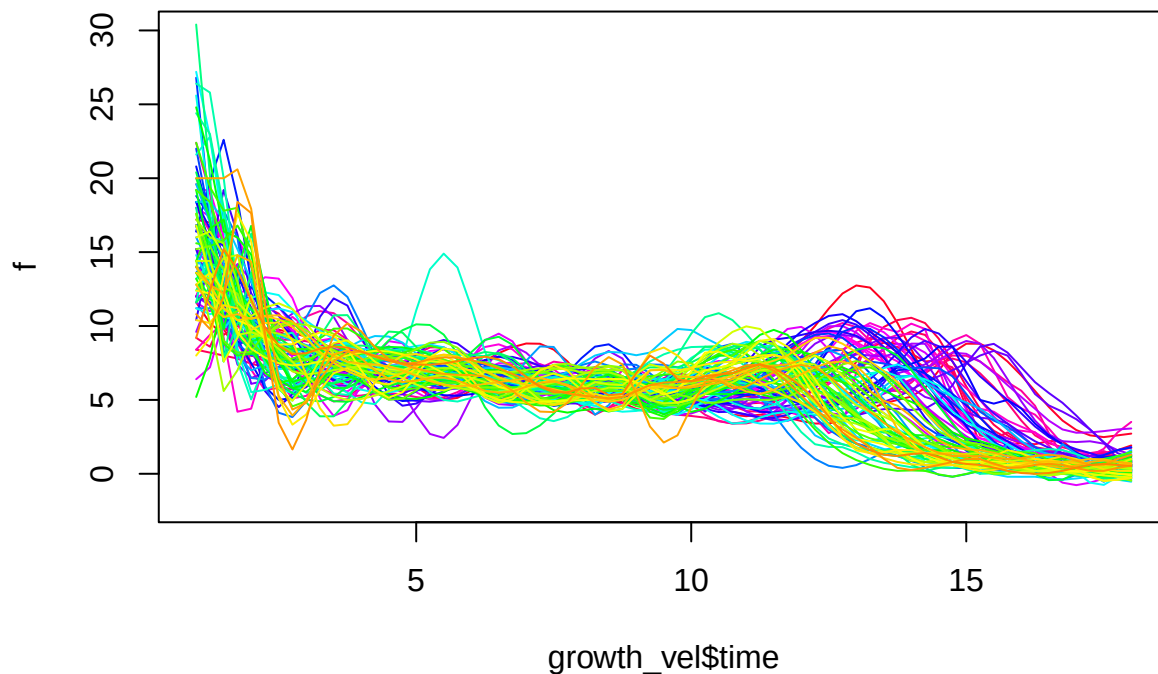
## $lambda
## [1] 68.25157
##
## $lambda.pvalue
## [1] 0.004975124
##
## $lambda.table.pvalue
## [1] 1.094358e-11
##
## $H
## [1] 6.355389
##
## $H.pvalue
## [1] 0.004975124
##
## $H.table.pvalue
```

```
## [1] 1.823731e-05
##
## $J
## [1] 59.21059
##
## $J.pvalue
## [1] 0.004975124
##
## $J.table.pvalue
## [1] 0
##
## $G
## [1] 8.647293
##
## $G.pvalue
## [1] 0.004975124
##
## $G.table.pvalue
## [1] 6.443113e-11
```

R Exercise 12: Fisher-Rao growth velocity data

Analyse growth velocity curves using Fisher-Rao registration and functional PCA tools.

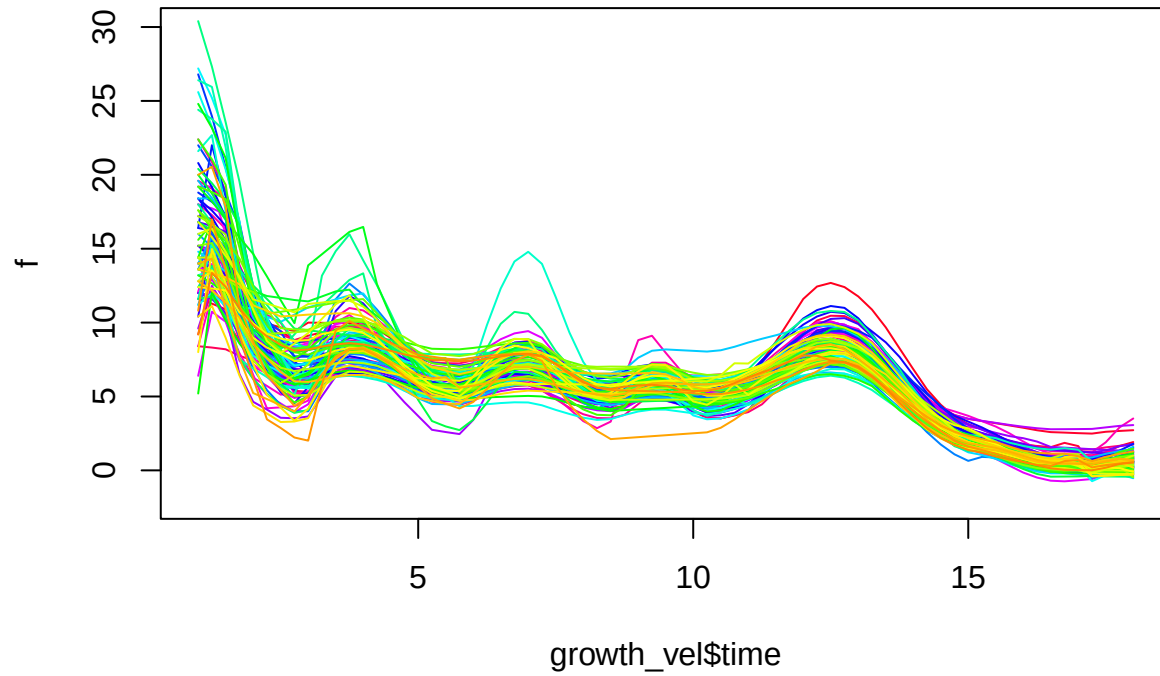
```
library("fdasrvf")
data(growth_vel)
n<-93
palette(rainbow(n+10))
plot(growth_vel$time,growth_vel$f[,1],ylim=c(-2,30),type="n",ylab="f")
for (i in 1:n){
  lines(growth_vel$time,growth_vel$f[,i],col=n+10-i)
}
```



```

out <- time_warping( growth_vel$f, growth_vel$time)
plot(growth_vel$time,growth_vel$f[,1],ylim=c(-2,30),type="n",ylab="f")
for (i in 1:n){
  lines(growth_vel$time,out$fn[,i],col=n+10-i)
}

```



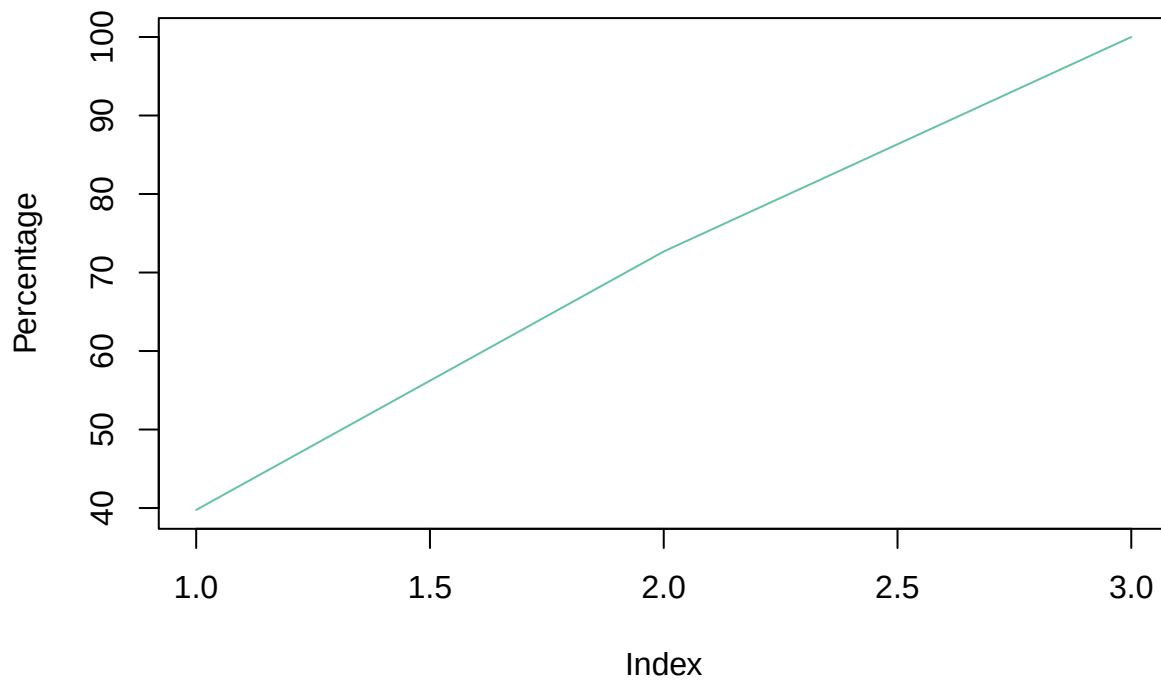
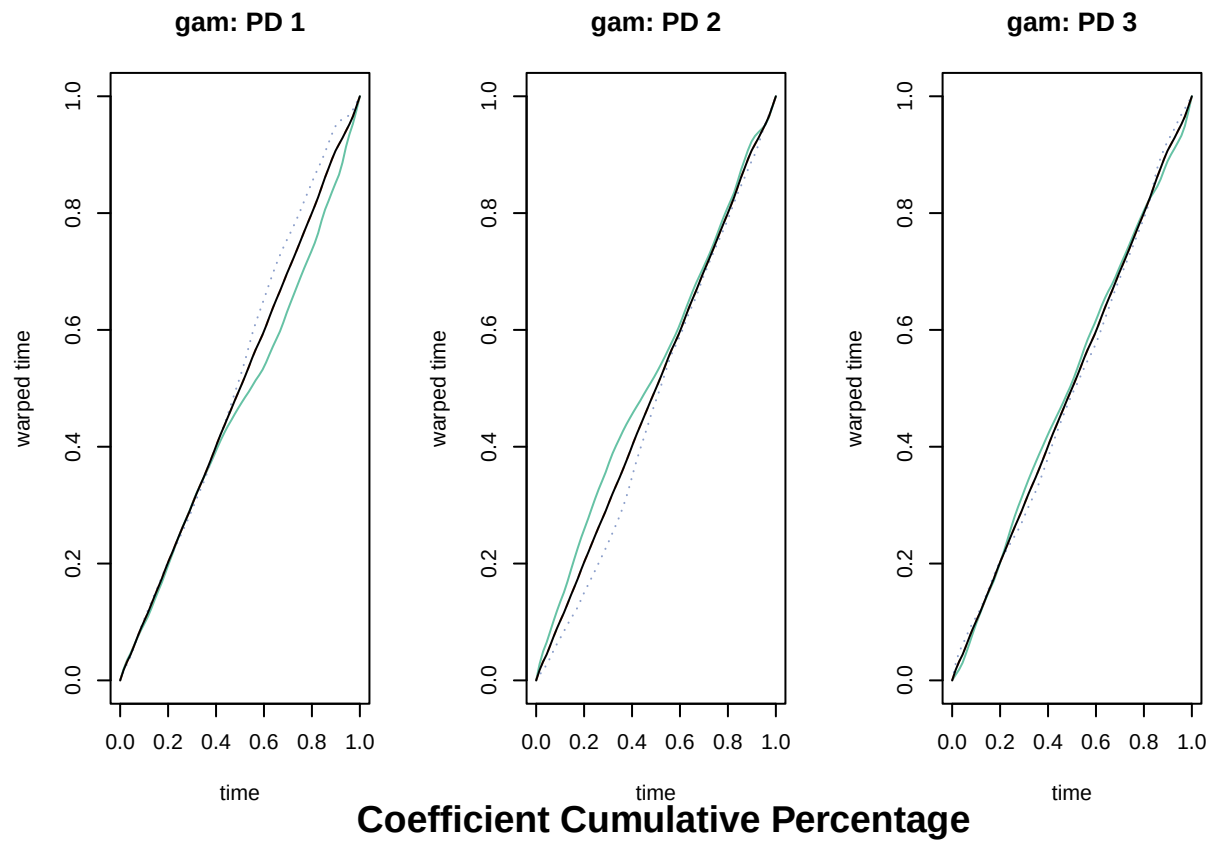
R Exercise 12: Fisher-Rao growth velocity (continued)

Analyse growth velocity curves using Fisher–Rao registration and functional PCA tools.

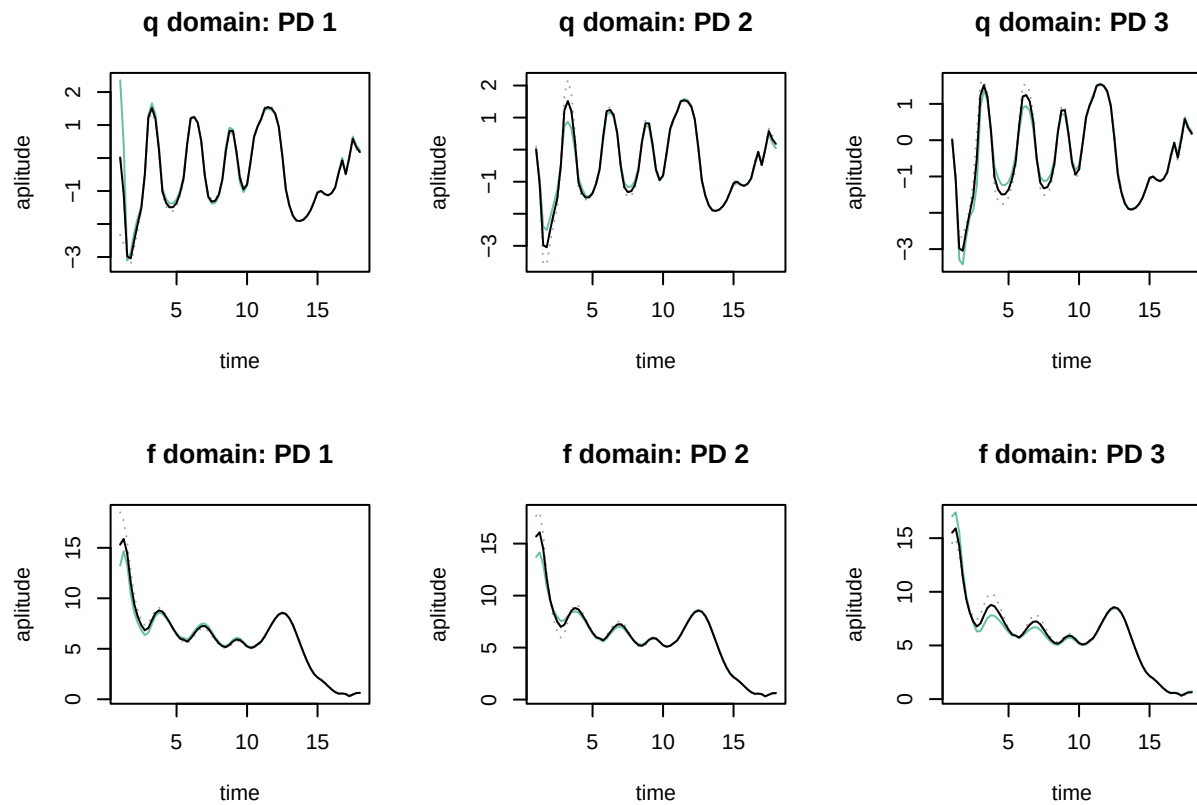
```

hfPCA <- horizFPCA(out, no = 3)

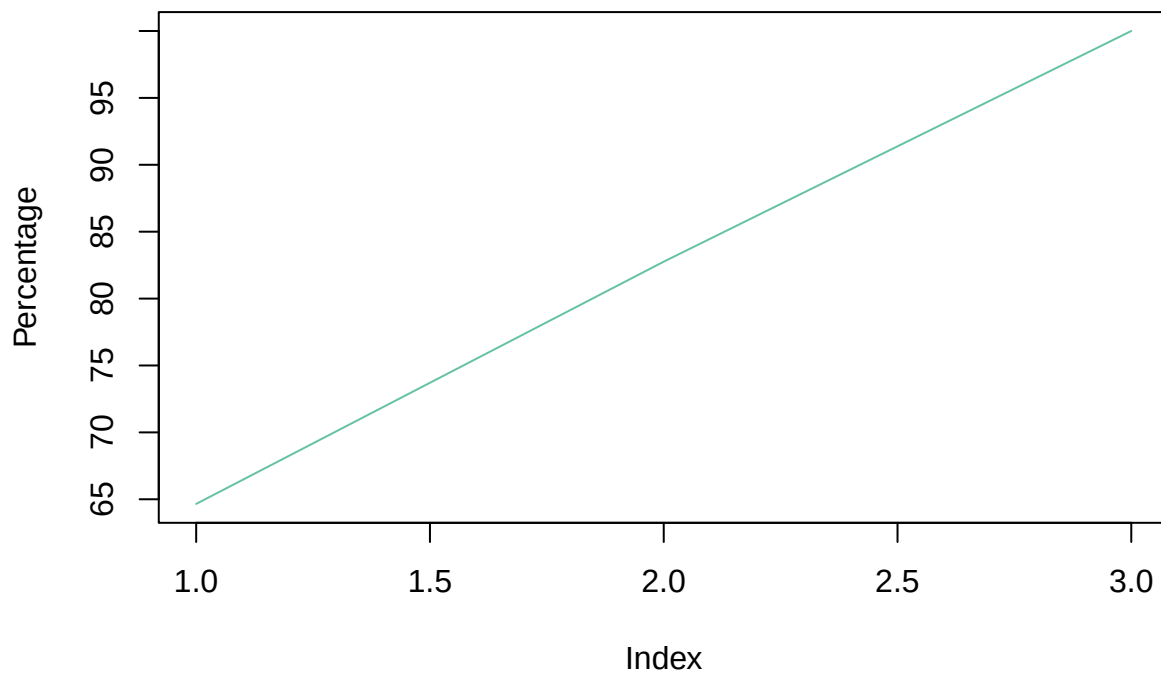
```



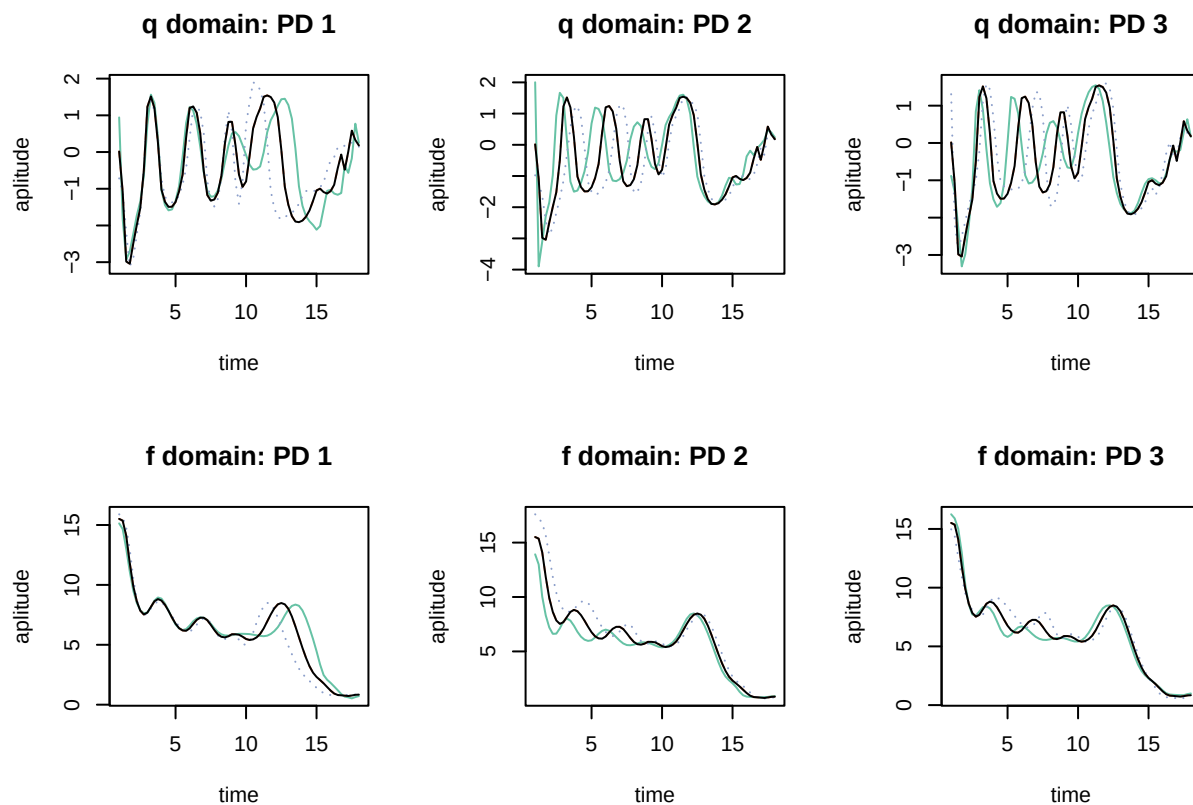
```
vfpc <- vertFPCA(out, no = 3)
```



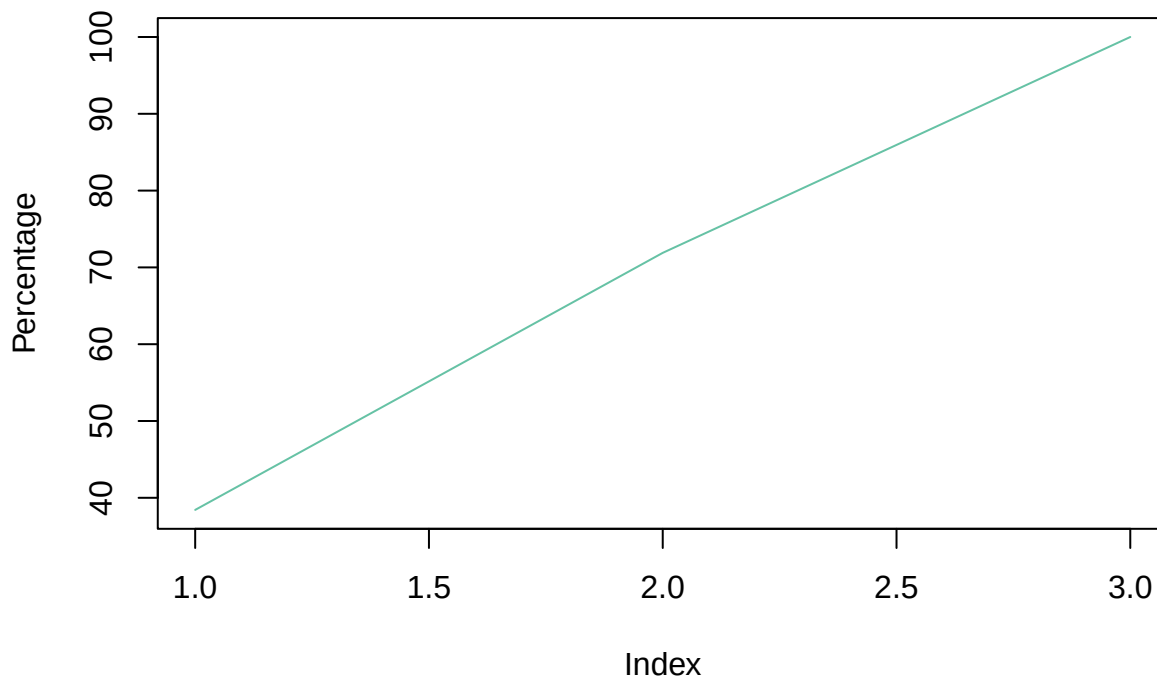
Coefficient Cumulative Percentage



```
jfpca <- jointFPCA(out, no = 3)
```

Coefficient Cumulative Percentage



R Exercise 13: Statistical shape analysis of six groups DNA data

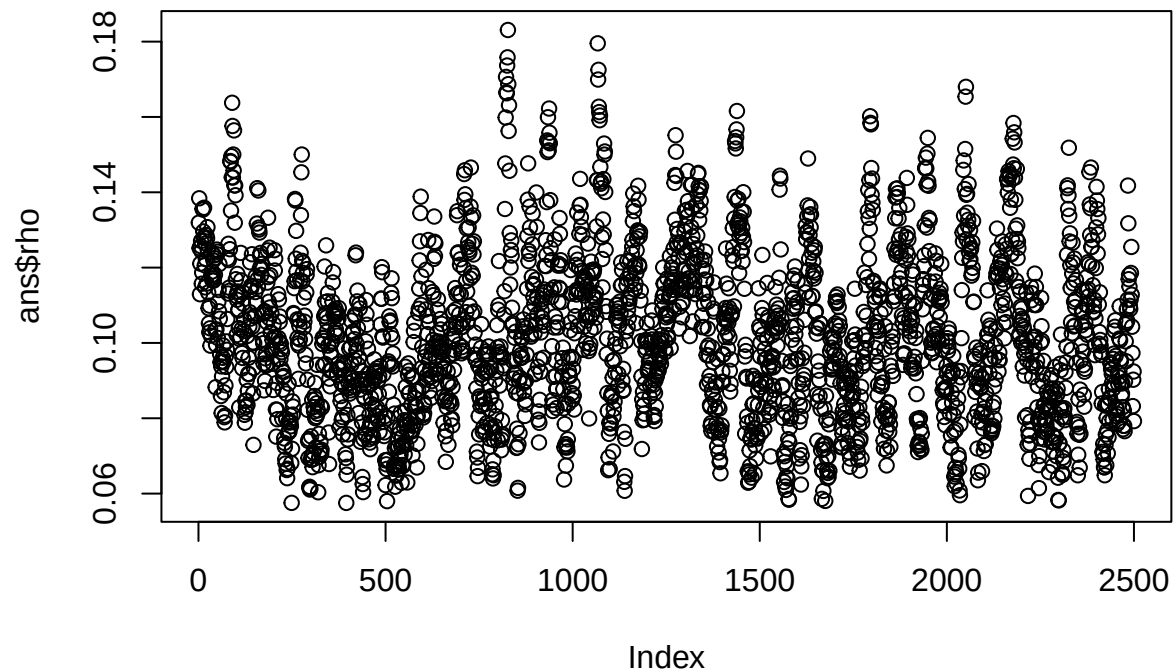
This R code reads in the data from 12 groups of DNA molecular dynamics simulations (undamaged XGX and damaged XFX) and then calculates a few simple summaries from the thinned AFA group.

```

library(shapes)
afa<-read.in("./DNA/AFA_P.x",22,3) ; afc<-read.in("./DNA/AFC_P.x",22,3)
afg<-read.in("./DNA/AFG_P.x",22,3) ; aga<-read.in("./DNA/AGA_P.x",22,3)
agc<-read.in("./DNA/AGC_P.x",22,3) ; agg<-read.in("./DNA/AGG_P.x",22,3)
tfa<-read.in("./DNA/TFA_P.x",22,3) ; tfg<-read.in("./DNA/TFC_P.x",22,3)
tft<-read.in("./DNA/TFT_P.x",22,3) ; tga<-read.in("./DNA/TGA_P.x",22,3)
tgc<-read.in("./DNA/TGC_P.x",22,3) ; tgt<-read.in("./DNA/TGT_P.x",22,3)

ans<-procGPA(afa,scale=FALSE) #investigate AFA group
shapes3d(ans$rotated)
plot(ans$rho) #shape distances `rho' from the mean shape

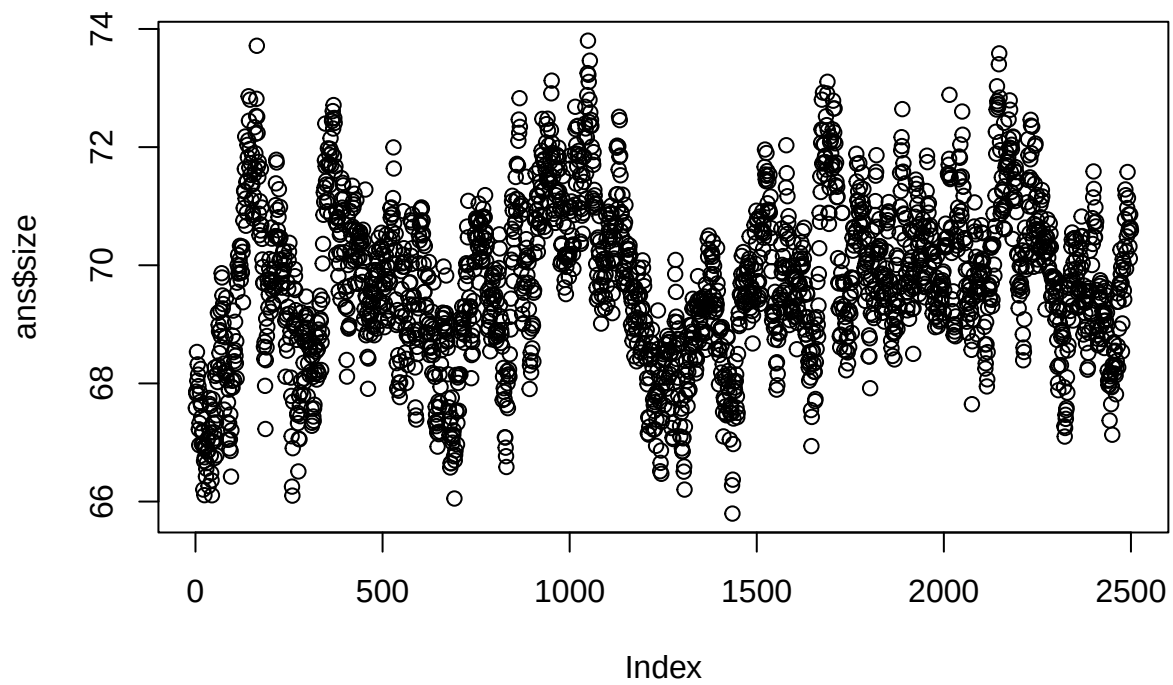
```



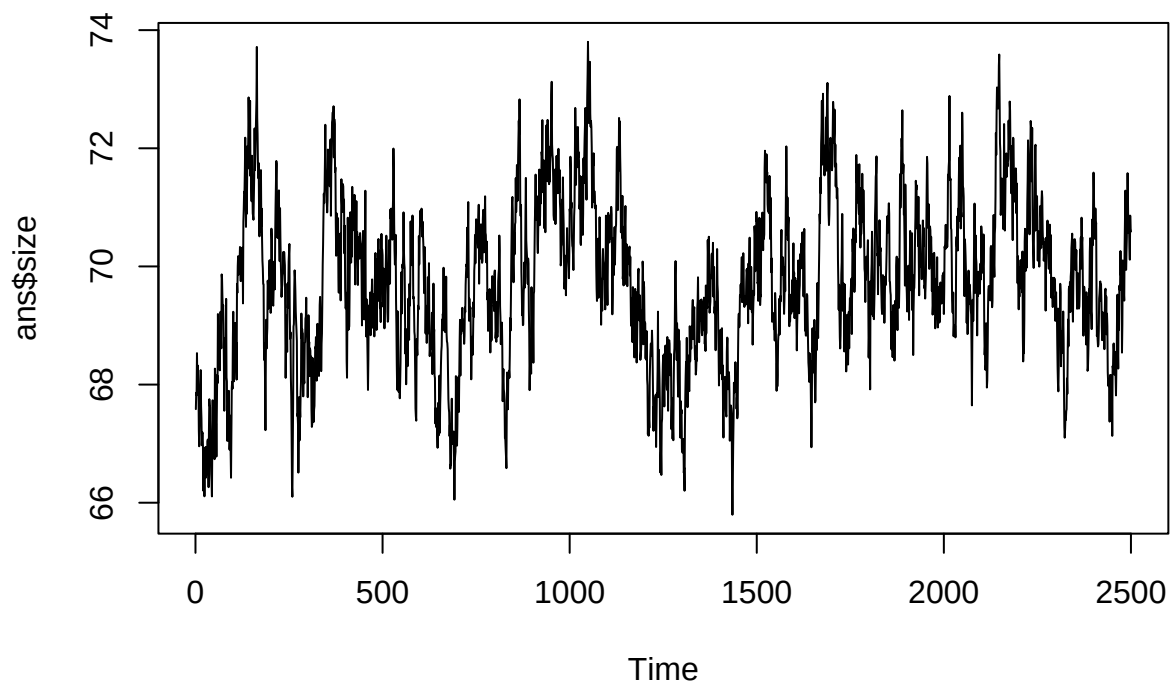
```

plot(ans$size) #size

```

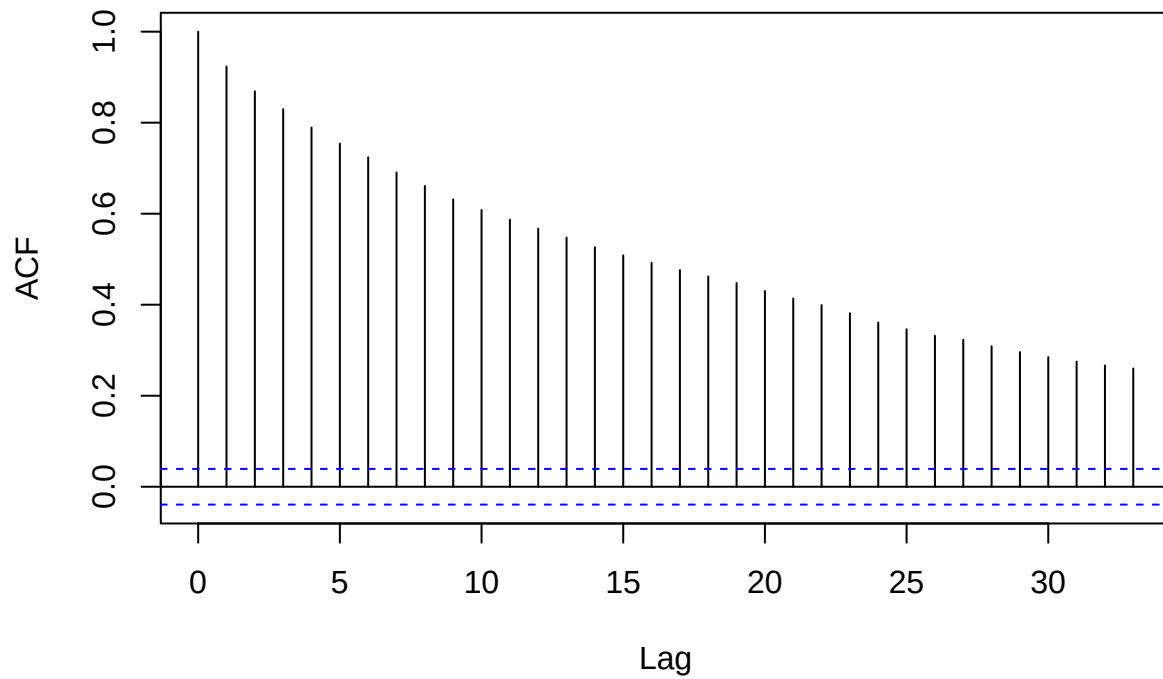


```
ts.plot(ans$size) # time series plot
```



```
acf(ans$size)
```

Series ans\$size



```
pacf(ans$size)
```

Series ans\$size

